

Supplementary Information

A Unified Ornstein–Uhlenbeck–Lévy–Branching Framework for Interpretable Modeling of Cancer Evolution

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Supplementary Note 1. Complete Mathematical Formulation

The Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework represents biological evolution as a latent stochastic process combining continuous constrained dynamics, discontinuous evolutionary transitions, lineage diversification, and heterogeneous observation processes within a unified probabilistic model.

Latent state

Let

$$X_t \in \mathbb{R}^q$$

denote the latent biological state at continuous time t .

The latent representation may correspond to

- pathway activities,
- ecological compositions,
- cell-state abundances,
- developmental coordinates,
- spatial ecological niches,
- latent embeddings learned from high-dimensional measurements.

The framework intentionally separates latent biological dynamics from measurement processes.

Continuous dynamics

Between abrupt evolutionary events,

$$dX_t = \Theta_t(\mu_t - X_t) dt + \Sigma_t dW_t,$$

where

- W_t is a multivariate Wiener process,
- Θ_t controls mean reversion,
- μ_t denotes the evolving attractor,
- Σ_t specifies stochastic diffusion.

When

$$\Theta_t = 0,$$

the model reduces to Brownian motion.

Therapy-dependent attractors

External perturbations may modify the equilibrium,

$$\mu_t = f(Z_t),$$

where Z_t denotes exogenous covariates that influence the latent evolutionary attractor. Examples include treatment, environmental conditions, or developmental programs.

Examples include

$$\mu_t = \mu_0 + \beta Z_t.$$

or nonlinear neural parameterizations.

Lévy jump process

Abrupt evolutionary events are modeled by

$$dJ_t = \int_{\mathbb{R}^q} z N(dt, dz),$$

where $N(dt, dz)$ is a Poisson random measure.

The latent dynamics become

$$dX_t = \Theta(\mu - X_t) dt + \Sigma dW_t + dJ_t.$$

Branching dynamics

Suppose $B_t \in \{1, \dots, K\}$ denotes the lineage identity.

Branch transitions follow

$$P(B_{t+\Delta} = j \mid B_t = i) = \Pi_{ij}.$$

Each branch may possess branch-specific parameters $(\Theta_k, \mu_k, \Sigma_k)$.

Observation model

Observed measurements Y_i are generated through

$$Y_i \sim p(Y_i \mid X_{t_i}, \phi).$$

Different likelihoods include

- Gaussian
- Poisson
- Negative Binomial
- Beta
- Dirichlet
- Multinomial
- Logistic-normal

without modifying latent dynamics.

Hierarchical Bayesian inference

Unknown parameters

$$\Psi = \{\mu, \Theta, \Sigma, \Pi, \phi, \lambda, \nu\}$$

are estimated through

$$p(\Psi, X, B, J \mid Y) \propto p(Y \mid X, \Psi) p(X, J, B \mid \Psi) p(\Psi).$$

Posterior inference is performed using Hamiltonian Monte Carlo.

General model

The complete OULB process is therefore

$$\boxed{dX_t = \Theta_{B_t,t}(\mu_{B_t,t} - X_t) dt + \Sigma_{B_t,t} dW_t + dJ_t, \quad Y_t \sim p(Y_t \mid X_t, \phi).}$$

Supplementary Note 2. Nested Special Cases

A major strength of the OULB framework is that many existing stochastic models arise as constrained special cases.

Model	OU	Lévy	Branching
Brownian motion	No	No	No
OU process	Yes	No	No
OU+Jump	Yes	Yes	No
OU+Branching	Yes	No	Yes
State-space model	Optional	Optional	Optional
Full OULB	Yes	Yes	Yes

Nested models within the OULB framework.

Supplementary Note 3. Bayesian Inference

Inference was performed using component-specific procedures appropriate to each validation experiment. Continuous OU parameters were estimated from the exact irregular-time transition likelihood. Jump events were evaluated using standardized OU innovations and a spike-and-slab-inspired event model. Branch assignments were reconstructed using probabilistic mixture models followed by optimal label alignment. Bootstrap resampling was used to quantify uncertainty in aggregate performance metrics. Hamiltonian Monte Carlo may be used for full Bayesian implementations of the general OULB model.

For fully Bayesian implementations based on Hamiltonian Monte Carlo, posterior convergence may be assessed using

- $\hat{R} < 1.01$,
- effective sample size,
- trace plots, and
- posterior predictive checks.

When Bayesian model comparison is required, candidate models may be evaluated using

- the widely applicable information criterion (WAIC),
- Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO), and
- Bayes factors, where appropriate.

Supplementary Note 4. Observation Models

The latent stochastic process is independent of measurement technology.

Examples include

- single-cell RNA sequencing,
- spatial transcriptomics,
- pathway activity estimation,
- ecological state composition,
- imaging-derived latent representations.

Observation likelihoods may therefore be replaced without altering the latent evolutionary process.

Supplementary Note 5. Cross-scale Interpretation

The latent state may represent biological organization at multiple levels,

Genes $\rightarrow$ Pathways $\rightarrow$ Cell States $\rightarrow$ Ecological States $\rightarrow$ Spatial Niches $\rightarrow$ Patient Evolution.

The OULB framework treats these representations identically from a probabilistic perspective.

Supplementary Note 6. Simulation Framework

Simulation studies evaluated

1. Brownian motion
2. OU dynamics
3. Therapy-dependent attractors
4. Lévy jumps
5. Branching evolution
6. Combined OULB dynamics

Simulation scenarios and activated model components are summarized in Supplementary Table S3.

Supplementary Note 7. Parameter Recovery

Simulation experiments quantified recovery of

- attractor location,
- mean reversion,
- diffusion,
- jump frequency,
- jump magnitude,
- branching probabilities.

Evaluation metrics included

- bias,
- RMSE,
- credible interval coverage,
- posterior calibration.

Supplementary Note 8. Sensitivity Analyses

Sensitivity analyses varied

- observation noise,
- irregular sampling intervals,
- missing observations,
- sample size,
- latent-state dimensionality,
- jump magnitude,
- branch separation or branching probability,
- prior specification, and
- computational runtime.

Supplementary Note 9. Computational Implementation

Software was implemented in Python using

- NumPy,
- SciPy,
- PyMC,
- ArviZ,
- pandas,
- scikit-learn.

Complete source code is available in the accompanying GitHub repository.

Supplementary Note 10. Illustrative Leukemia Application

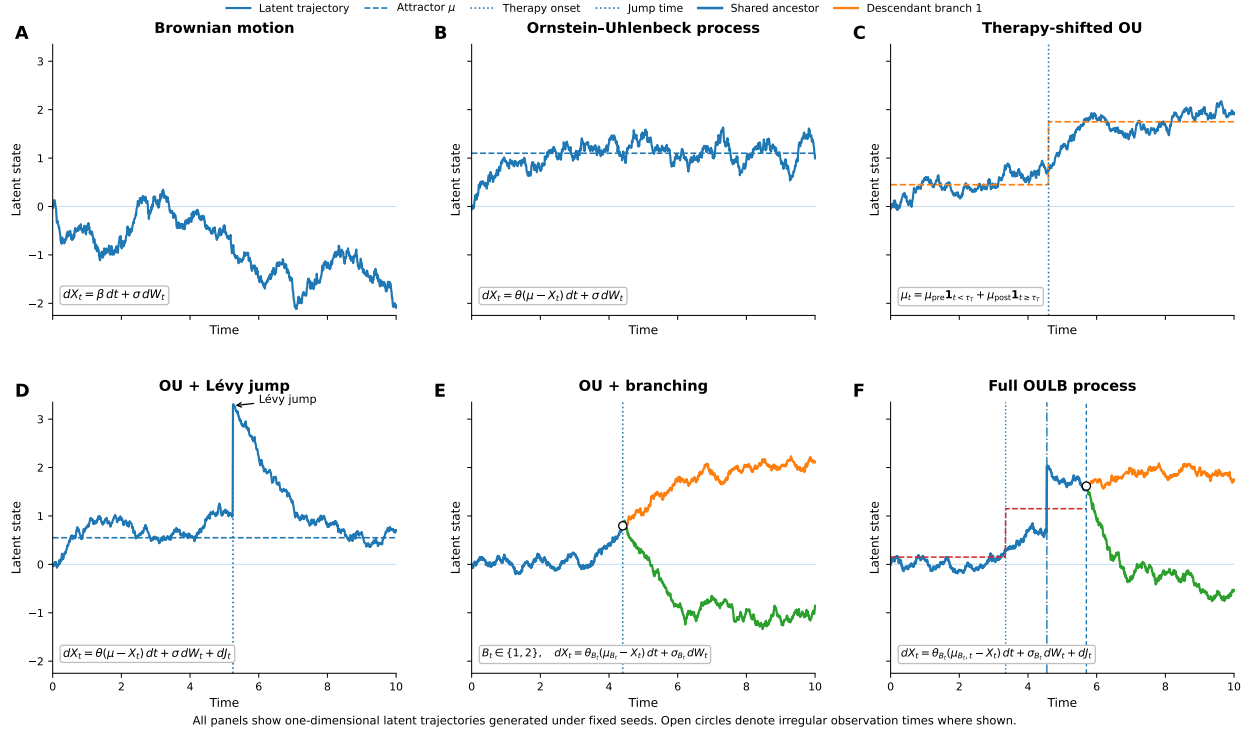
The pediatric leukemia application demonstrates one realization of the general OULB framework.

The framework itself is disease independent and may be applied to developmental biology, microbial evolution, ecological dynamics, phylogenetics, and other biological systems.

Supplementary Figures

The following supplementary figures provide systematic validation of the Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework. Supplementary Figure S1 illustrates the representative stochastic regimes encompassed by the model. Supplementary Figures S2–S4 evaluate recovery of the continuous OU parameters, detection of Lévy jumps, and reconstruction of branching structure, respectively. Supplementary Figure S5 examines joint recovery when continuous dynamics, abrupt jumps, and lineage branching operate simultaneously, whereas Supplementary Figure S6 evaluates robustness to observation noise, irregular sampling, missing data, sample size, latent-state dimensionality, and computational scaling.

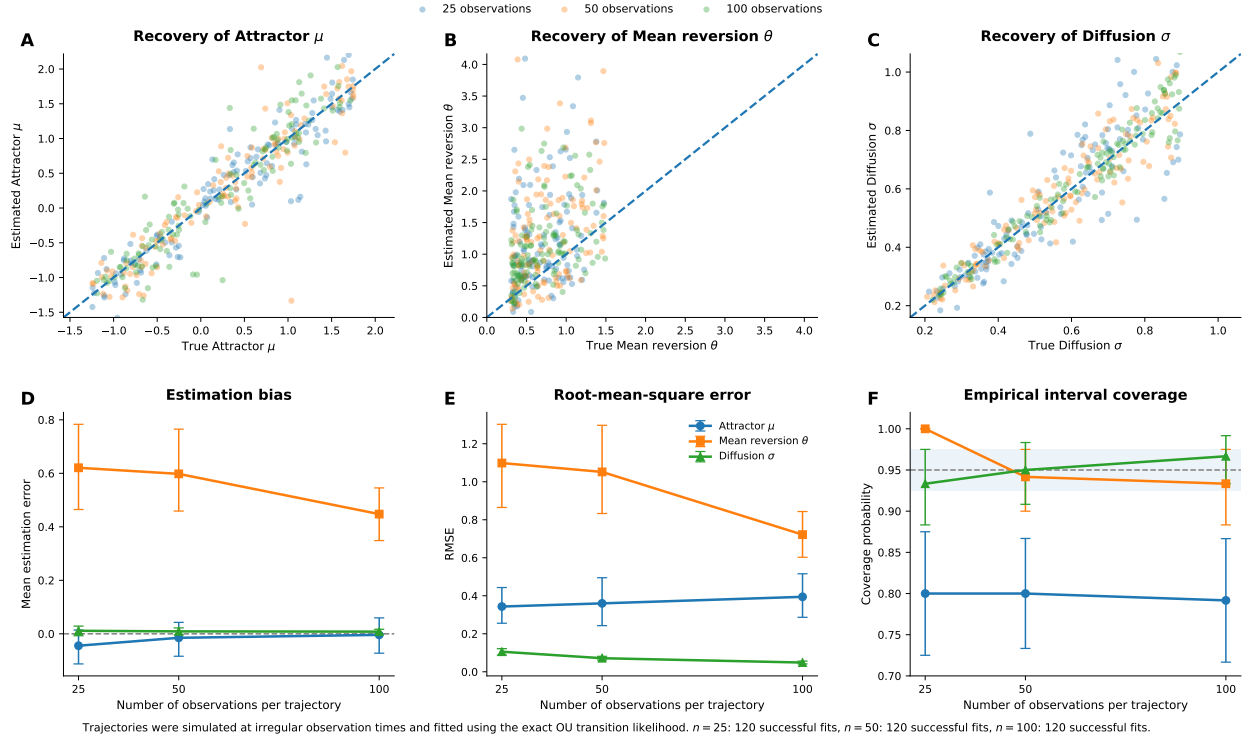
Supplementary Figure S1. Representative simulated stochastic regimes



Supplementary Figure S1. Representative stochastic regimes generated by the OULB framework.

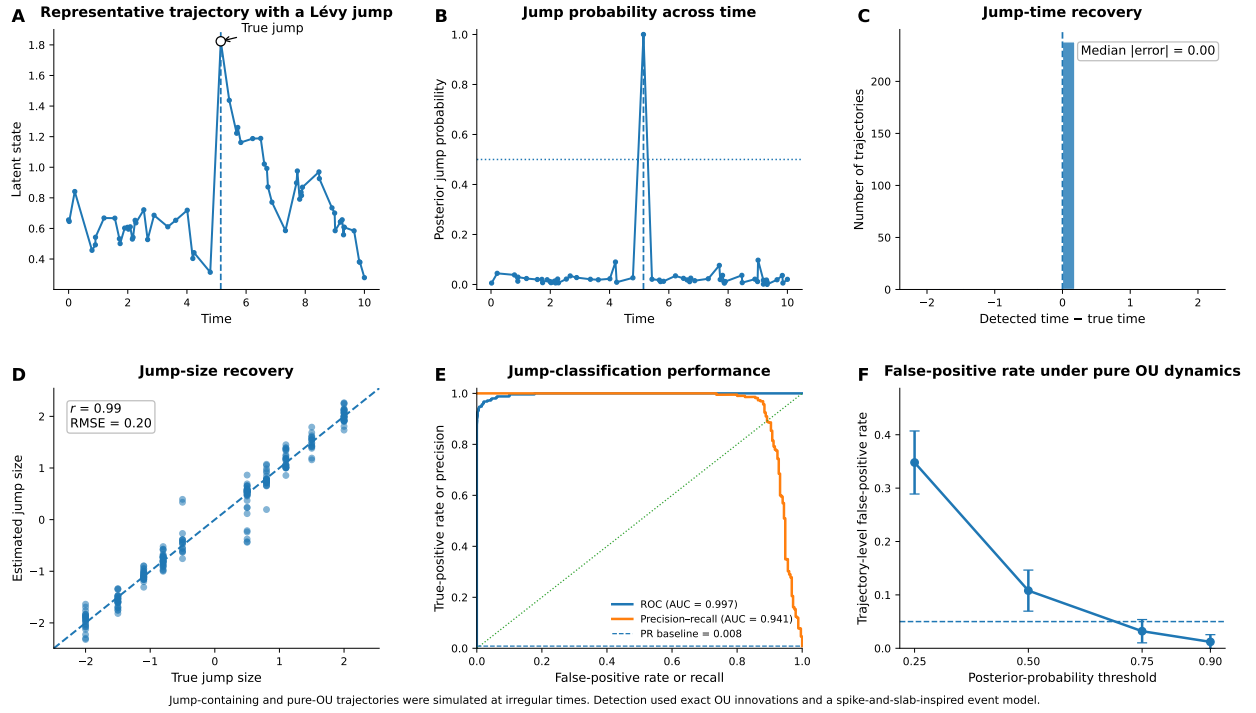
Representative one-dimensional trajectories illustrating the principal stochastic behaviors encompassed by the Ornstein–Uhlenbeck–Lévy–Branching (OULB) framework. (A) Brownian motion exhibiting unconstrained diffusion. (B) Ornstein–Uhlenbeck (OU) dynamics with mean reversion toward a stable attractor. (C) Therapy-dependent OU dynamics following an abrupt attractor shift. (D) OU dynamics with a single Lévy jump. (E) OU dynamics with lineage branching into distinct descendant trajectories. (F) Integrated OULB dynamics combining continuous OU evolution, a Lévy jump, and lineage branching within a unified stochastic process.

Supplementary Figure S2. Recovery of Ornstein-Uhlenbeck parameters



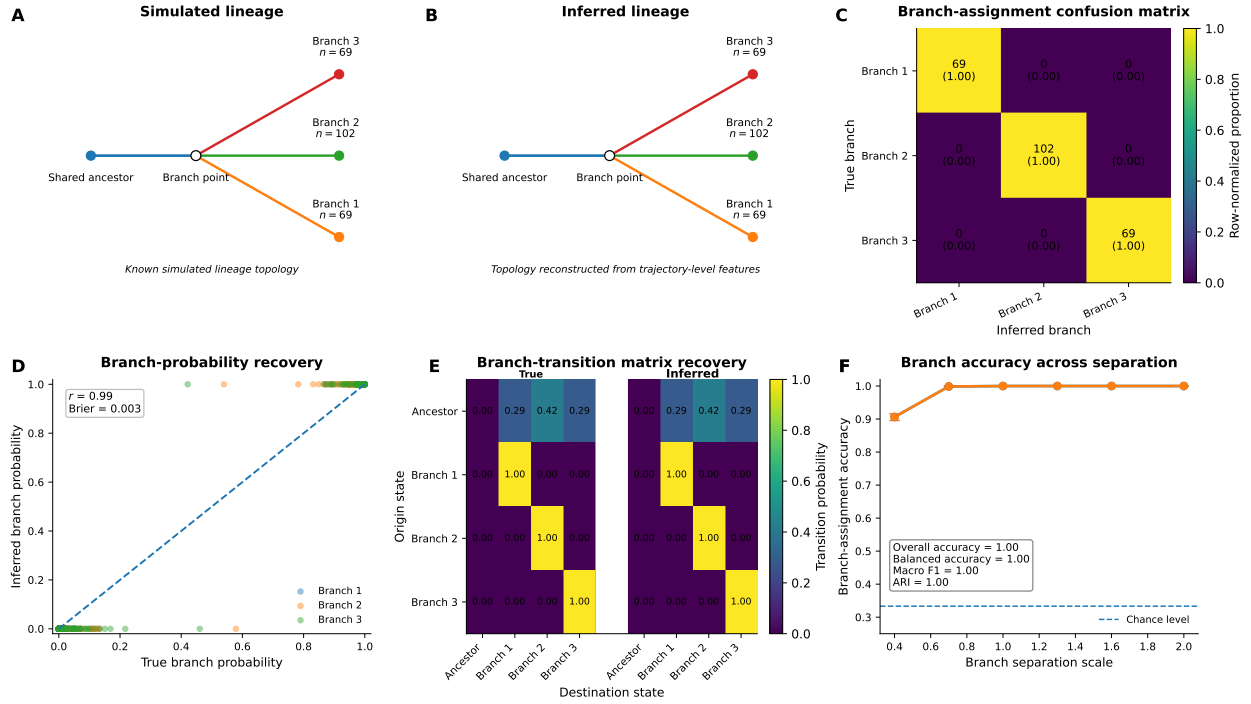
Supplementary Figure S2. Recovery of Ornstein-Uhlenbeck parameters. Simulation-based evaluation of parameter recovery under irregular longitudinal sampling. (A) Recovery of the attractor parameter (μ). (B) Recovery of the mean-reversion parameter (θ). (C) Recovery of the diffusion parameter (σ). (D) Mean estimation bias as a function of the number of observations per trajectory. (E) Root-mean-square error (RMSE) across sampling densities. (F) Empirical 95% interval coverage. Points in panels A–C represent individual simulation replicates, and diagonal dashed lines denote identity. Lines in panels D–F represent mean performance across simulation replicates; error bars denote bootstrap 95% confidence intervals.

Supplementary Figure S3. Validation of Lévy jump detection



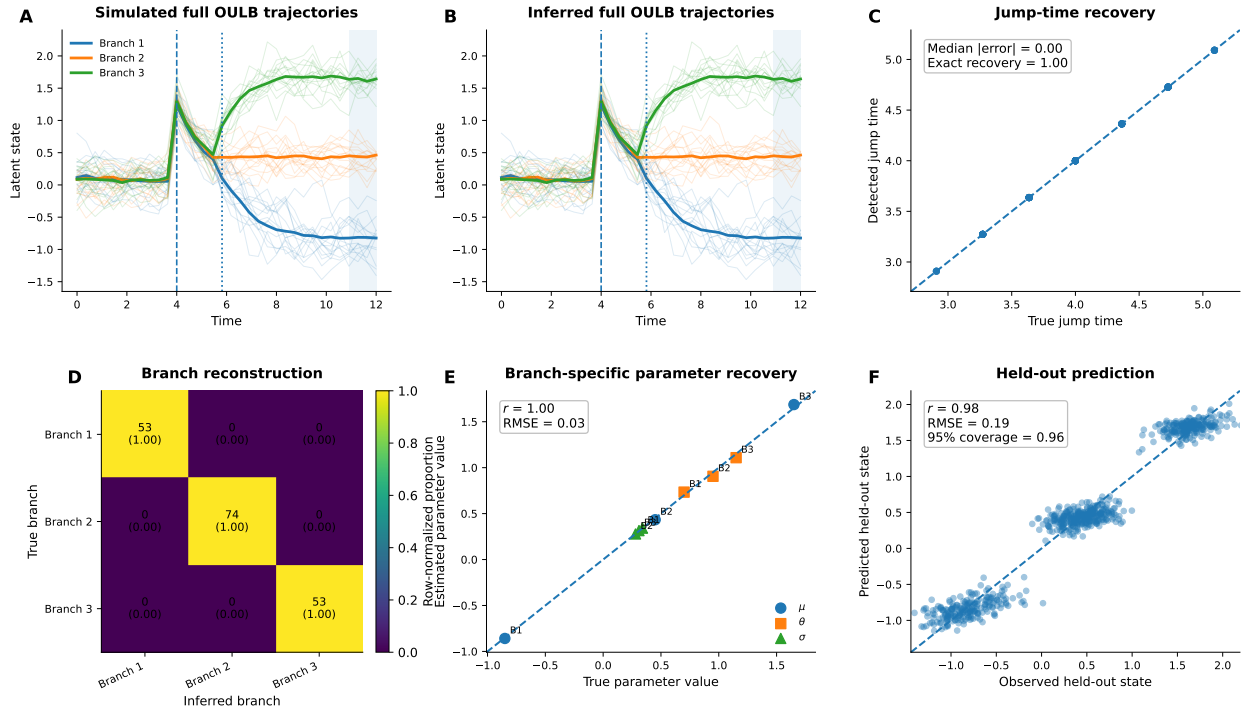
Supplementary Figure S3. Validation of Lévy jump detection. Simulation-based validation of the Lévy jump detection component of the OULB framework. (A) Representative latent trajectory containing a single simulated Lévy jump. (B) Posterior jump probability accurately localizes the discontinuity. (C) Distribution of jump-time recovery errors across simulated trajectories. (D) Recovery of jump magnitude demonstrating strong agreement between true and estimated values. (E) Receiver operating characteristic (ROC) and precision–recall (PR) analyses demonstrating high discrimination performance. (F) Trajectory-level false-positive rate as a function of the posterior jump-probability threshold under pure OU dynamics.

Supplementary Figure S4. Validation of branch reconstruction



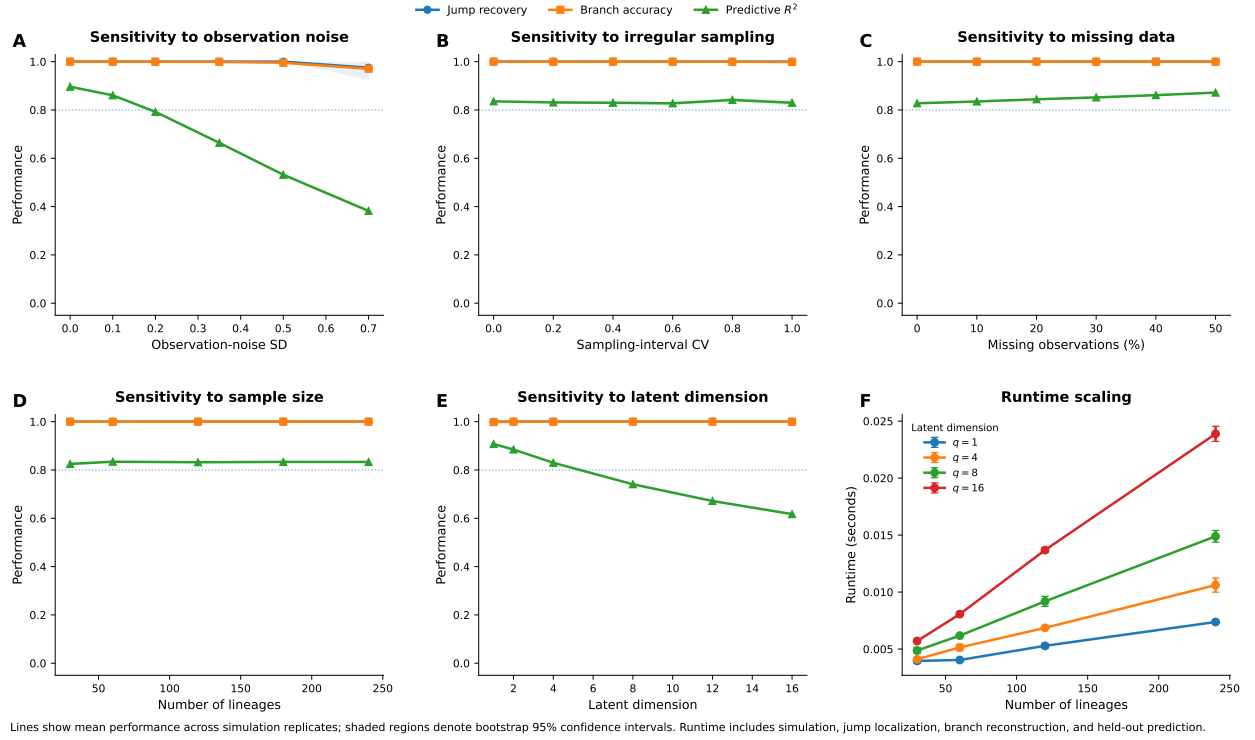
Supplementary Figure S4. Validation of branch reconstruction. Simulation-based validation of lineage reconstruction within the OULB framework. (A) Simulated lineage tree with three descendant branches. (B) Reconstructed lineage inferred from trajectory-level features. (C) Branch-assignment confusion matrix following optimal label alignment. (D) Recovery of branch membership probabilities. (E) Comparison of true and inferred branch-transition matrices. (F) Branch-assignment accuracy as a function of branch separation.

Supplementary Figure S5. Joint recovery of the complete OULB model



Supplementary Figure S5. Joint recovery of the complete OULB model. Integrated validation of simultaneous recovery of continuous dynamics, discontinuous events, and branching structure. (A) Simulated trajectories generated under the complete OULB model, including continuous OU dynamics, a shared Lévy jump, lineage branching, and branch-specific stochastic dynamics. (B) Corresponding inferred trajectories. (C) Recovery of the shared jump time. (D) Branch-assignment confusion matrix. (E) Recovery of branch-specific OU parameters (μ , θ , and σ). (F) Prediction of held-out observations using the inferred OULB model.

Supplementary Figure S6. Sensitivity analyses of the OULB framework



Supplementary Figure S6. Sensitivity analyses of the OULB framework. Robustness of the integrated OULB framework under progressively more challenging data conditions. Performance was evaluated with increasing (A) observation noise, (B) irregular sampling, (C) missing observations, (D) varying sample size, and (E) latent-state dimensionality. (F) Computational runtime as a function of sample size and latent dimension. Lines denote mean performance across simulation replicates, and shaded regions represent bootstrap 95% confidence intervals.

Supplementary Tables

The following supplementary tables provide reference information and numerical summaries supporting the mathematical formulation, simulation experiments, and validation analyses. The tables define the notation and prior distributions used in the OULB framework, summarize the simulated stochastic regimes, and report quantitative results for OU parameter recovery, Lévy jump detection, branch reconstruction, joint OULB recovery, and sensitivity analyses.

Supplementary Table S1. Mathematical notation used throughout the OULB framework.

Symbol	Description	Dimension	Units
X_t	Latent biological state	q	arbitrary
Y_t	Observed measurement	variable	measurement-specific
μ	OU attractor	q	latent-state units
θ	Mean-reversion parameter	scalar or vector	time ⁻¹
σ	Diffusion parameter	scalar or vector	latent-state units · time ^{-1/2}
W_t	Wiener process	q	—
J_t	Lévy jump process	q	latent-state units
B_t	Branch identity	discrete	—
Π	Branch-transition matrix	$K \times K$	probability
ϕ	Observation-model parameters	variable	model-specific

Supplementary Table S2. Prior distributions used for Bayesian inference.

Parameter	Prior	Hyperparameters
μ	Normal	$\mathcal{N}(0, 2^2)$
θ	Half-normal	scale = 1
σ	Half-normal	scale = 1
Jump rate λ	Exponential	rate = 1
Jump magnitude	Normal	$\mathcal{N}(0, \tau^2)$
Branch probabilities	Dirichlet	$\boldsymbol{\alpha} = (1, \dots, 1)$
Observation parameters ϕ	Model dependent	See Supplementary Note 4

Supplementary Table S3. Simulation scenarios used throughout the validation studies.

Simulation	Purpose	Components activated
Brownian motion	Unconstrained diffusion	Brownian
OU process	Continuous dynamics	OU
Therapy-shifted OU	Dynamic attractor	OU + therapy
OU + Lévy jump	Discontinuous evolution	OU + jump
OU + branching	Lineage diversification	OU + branching
Full OULB	Complete framework	OU + jump + branching

Supplementary Table S4. Summary of Ornstein–Uhlenbeck parameter recovery (Supplementary Figure S2).

Parameter	Observations	Mean bias	RMSE	95% coverage
Attractor (μ)	25	−0.044	0.343	0.800
Attractor (μ)	50	−0.015	0.360	0.800
Attractor (μ)	100	−0.004	0.394	0.792
Mean reversion (θ)	25	0.621	1.099	1.000
Mean reversion (θ)	50	0.598	1.052	0.942
Mean reversion (θ)	100	0.448	0.722	0.933
Diffusion (σ)	25	0.011	0.106	0.933
Diffusion (σ)	50	0.009	0.071	0.950
Diffusion (σ)	100	0.008	0.049	0.967

Supplementary Table S5. Summary of Lévy jump detection performance (Supplementary Figure S3).

Metric	Value
Median absolute jump-time error	0.000
Jump-size correlation	0.988
Jump-size RMSE	0.200
ROC AUC	0.997
Precision–recall AUC	0.941
False-positive rate at posterior threshold 0.50	0.108

Supplementary Table S6. Summary of branch reconstruction performance (Supplementary Figure S4).

Metric	Value
Overall accuracy	1.000
Balanced accuracy	1.000
Macro-F1 score	1.000
Adjusted Rand index	1.000
Mean Brier score	1.37×10^{-24}

Supplementary Table S7. Joint recovery performance of the complete OULB model (Supplementary Figure S5).

Component	Metric	Value
Jump recovery	Median absolute jump-time error	0.000
Jump recovery	Exact jump-time recovery	1.000
Branch reconstruction	Overall accuracy	1.000
Branch reconstruction	Balanced accuracy	1.000
Branch reconstruction	Macro-F1	1.000
Branch reconstruction	Adjusted Rand index	1.000
Parameter recovery	Correlation	0.999
Parameter recovery	RMSE	0.027
Prediction	Correlation	0.981
Prediction	RMSE	0.191
Prediction	95% interval coverage	0.960

Supplementary Table S8. Summary of sensitivity analyses (Supplementary Figure S6).

Perturbation	Jump recovery range	Branch accuracy range	Predictive R^2 range
Observation noise	0.975–1.000	0.971–1.000	0.382–0.896
Irregular sampling	1.000–1.000	0.999–1.000	0.827–0.842
Missing observations	1.000–1.000	1.000–1.000	0.828–0.872
Sample size	1.000–1.000	1.000–1.000	0.825–0.834
Latent dimension	1.000–1.000	0.998–1.000	0.617–0.907